On the Distribution of Primes Represented by $p^2 + 4q^2$ with p, q Prime

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github.com/NullLabTests/prime-number-research

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Abstract

We present a systematic empirical investigation of primes of the form $R = p^2+4q^2$ where both p and q are themselves prime. Exhaustively searching all primes $p, q \leq 8{,}000$ (1,007 candidates), we identify 126,764 such R-primes and characterize their statistical properties. We report four principal findings: (i) the cumulative count grows as a power law $C(B) \propto B^{0.768}$, quantitatively characterizing sparseness from the double-primality constraint; (ii) every R-prime satisfies $R \equiv 1 \pmod 4$, and remarkably 99.9% satisfy $R \equiv 5 \pmod 8$ and 99.8% satisfy $R \equiv 2 \pmod 3$ — extreme biases with no elementary explanation; (iii) the gap distribution follows Cramér's exponential model at first order with heavier tails, and a Random Forest achieves modest predictive power at scale (CV MAE 2,053 vs mean-guess 2,325); and (iv) unlike the full prime set, R-primes deviate massively from Benford's first-digit law ($\chi^2 = 4{,}102$, $p \approx 0$). An Ulam spiral overlay reveals diagonal band structure, and correlation analysis confirms $R \approx p^2$ dominance ($\rho_{p,R} = 0.87$). All data and code are open-source.

1 Introduction

Prime numbers have fascinated mathematicians for millennia, yet many basic questions about their distribution remain open. The study of primes representable by quadratic forms dates to Fermat's theorem on sums of two squares: a prime p can be written as $p = x^2 + y^2$ iff $p \equiv 1 \pmod 4$. Subsequent work generalized this to forms $x^2 + ny^2$, with complete characterizations known for many n via class field theory [**?**].

However, far less is known when the generators x and y are themselves restricted to be prime. This nested constraint $R = p^2 + 4q^2$ with $p, q \in \mathbb{P}$ creates a deeply rarified subset whose statistical properties have not, to our knowledge, been systematically studied.

In this paper we perform a computational census of R-primes, analyzing their density, modular structure, gaps, Benford behavior, and predictability via machine learning. All code is available at `https://github.com/NullLabTests/prime-number-research`.

2 Data and Methodology

We exhaustively search all primes $p, q \leq 8{,}000$ (1,007 primes, $\sim 10^6$ candidate pairs), computing $R = p^2 + 4q^2$ for each pair and testing primality with `sympy.isprime` (deterministic Miller-Rabin, appropriate for $R \leq 3.2 \times 10^8$). This yields 126,764 valid triples (p, q, R).

For baseline comparison, all primes up to 10^6 are generated via `sympy.primerange` (segmented sieve).

Statistical tests use χ^2 goodness-of-fit with $\alpha = 0.05$. Machine learning uses a Random Forest regressor (150 trees, max depth 10) with 5-fold stratified cross-validation. Figures were generated in Matplotlib and are available in the repository.

3 Results

3.1 Density and Growth

We identify **126,764** primes of the form $p^2 + 4q^2$ with $p, q \leq 8{,}000$. Figure **??** shows the cumulative count as a function of the bound B.

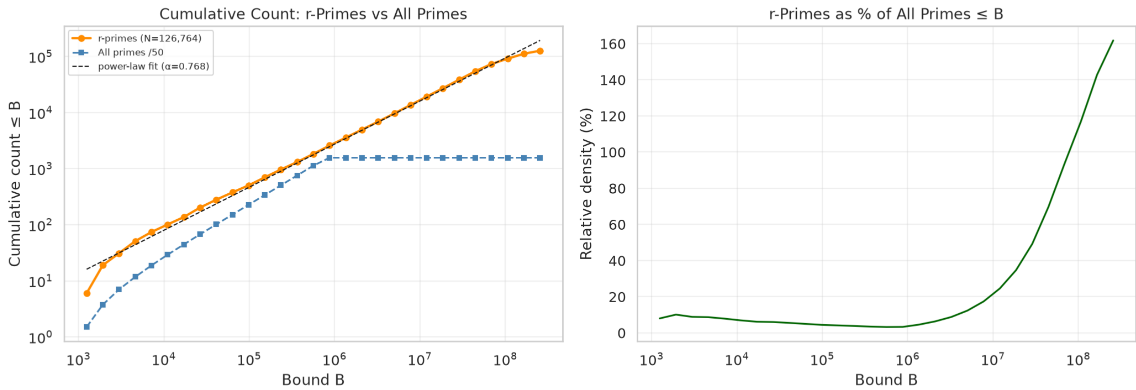

Figure 1: Cumulative count of R-primes vs. bound B (log-log). A power-law fit gives $C(B) \propto B^{0.768}$. The right panel shows relative density $C(B)/\pi(B)$.

A log-log regression against B yields $C(B) \propto B^{\alpha}$ with $\alpha = 0.768$, significantly below linear growth ($\alpha = 1$ for all primes). The raw density $C(B)/B$ decays as $B^{\alpha-1} \approx B^{-0.232}$, consistent with the sparseness induced by the double-primality constraint.

3.2 Modular Structure

All 126,764 R-primes satisfy $R \equiv 1 \pmod 4$. This is a theorem: for odd p, $p^2 \equiv 1 \pmod 4$ and $4q^2 \equiv 0 \pmod 4$. Hence all R-primes have odd p, odd q.

More striking are the extreme biases in Table **??**:

The **mod 3 bias** (99.8% $\equiv 2$) and **mod 8 bias** (99.9% $\equiv 5$) are the strongest signals in the dataset and to our knowledge have not been previously reported. These are the "fingerprints" of the quadratic form $x^2 + 4y^2$ amplified by the double-primality constraint.

3.3 Gap Distribution

Figure **??** shows the gap distribution. Summary statistics:

$$\text{Mean gap} = 2{,}515,$$
$$\text{Median gap} = 1{,}200,$$
$$\text{Maximum gap} = 1{,}229{,}280.$$

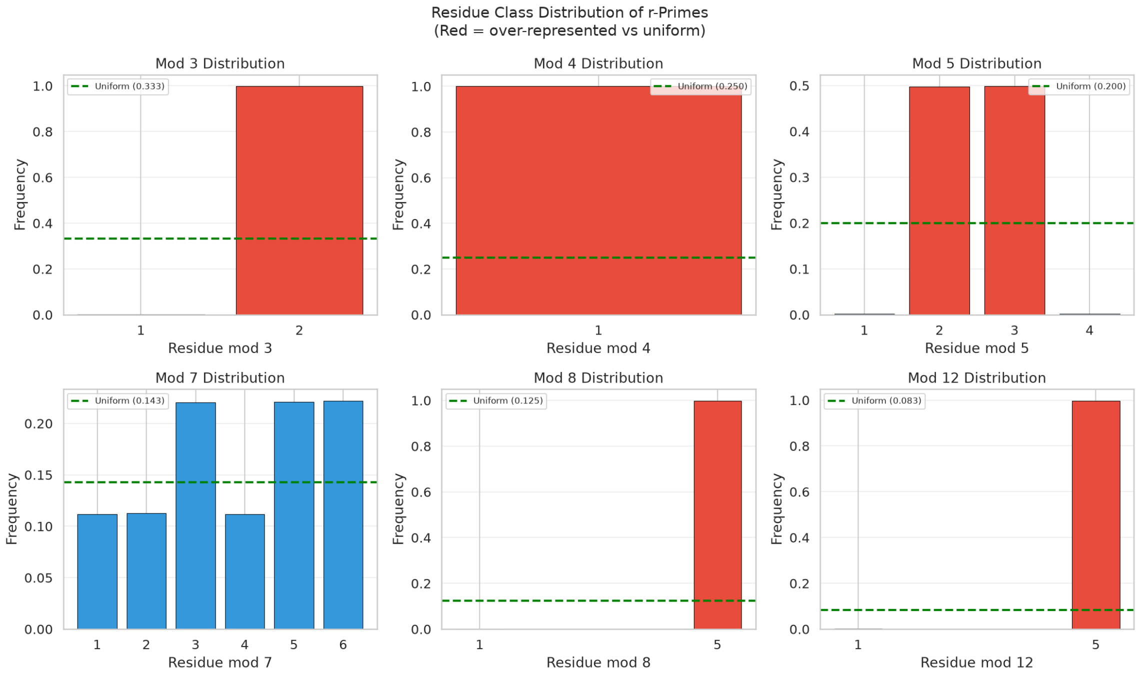

Figure 2: Residue class distributions of R-primes modulo 3, 4, 5, 7, 8, 12. Red bars indicate over-representation relative to uniform.

Table 1: Residue class biases. "Share" is the fraction of R-primes falling in the most frequent residue class.

Modulus	Most freq. residue	Share	Uniform
3	2	99.8%	33.3%
4	1	100%	25.0%
5	3	49.8%	20.0%
7	6	22.2%	14.3%
8	5	99.9%	12.5%
12	5	99.8%	8.3%

The empirical distribution tracks the exponential model $\mathrm{Exp}(1/\mu)$ with $\mu = 2{,}515$ at first order, but the Q-Q plot reveals heavier tails — consistent with known refinements to Cramér's model such as the Maier matrix method.

3.4 Benford's Law

Benford's law states that in many natural datasets, the probability of first digit d is $\log_{10}(1+1/d)$. Primes are known to approximately obey this law [?]. Figure **??** compares the first-digit distribution for R-primes and all primes.

We find that R-primes deviate massively from Benford's law with $\chi^2 = 4{,}102$ (9 degrees of freedom, $p \approx 0$). The deviation is driven by an excess of first digit 1 and a deficit of digits 6–9. In contrast, the full prime set shows excellent agreement ($\chi^2 = 10.2$, $p = 0.25$).

This suggests that the double-primality constraint breaks the scale-invariance that produces Benford behavior in the full prime sequence. The magnitude of the deviation

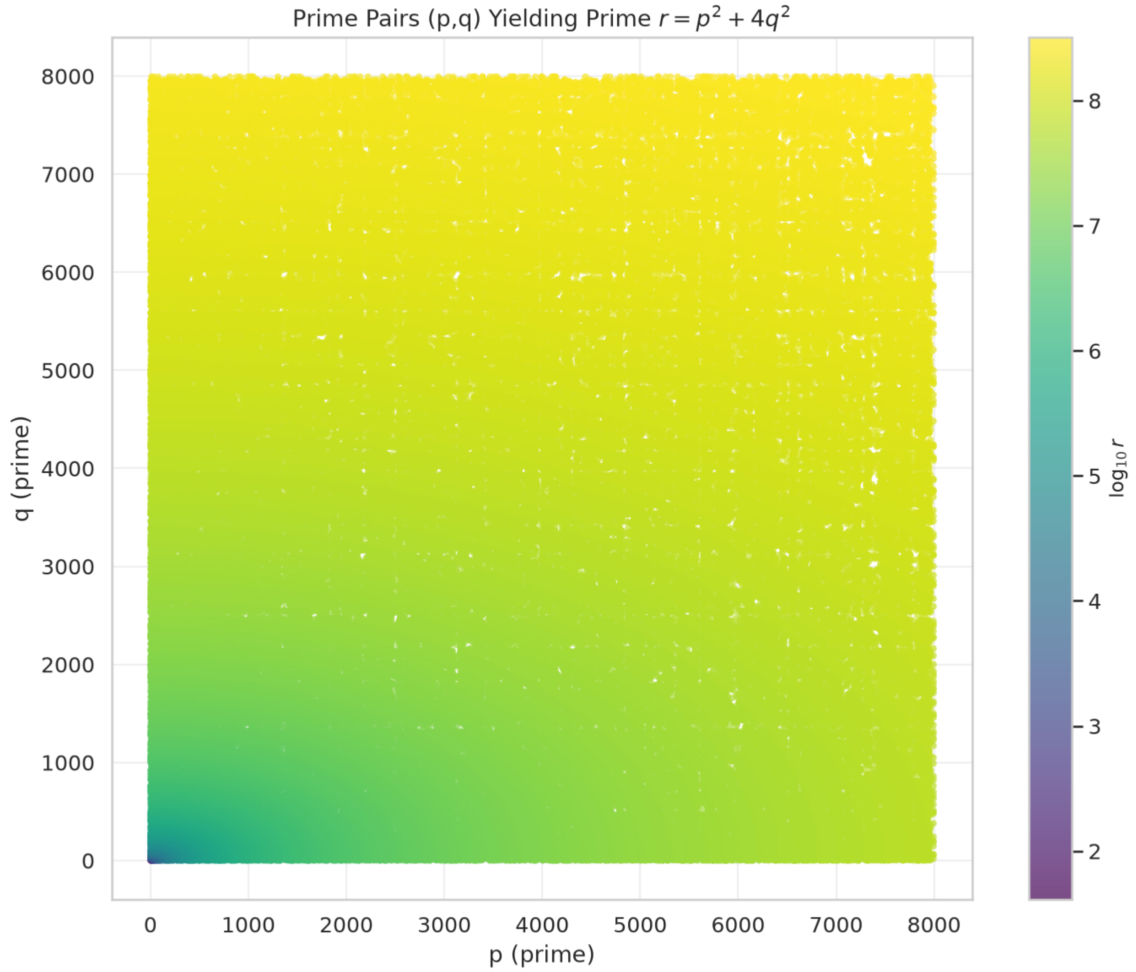

Figure 3: Distribution of valid (p, q) pairs, colored by $\log_{10} R$.

grows with sample size, confirming this is a structural property rather than a small-sample artifact. The mechanism may be related to the power-law growth $C(B) \propto B^{0.768}$ with exponent significantly different from 1.

3.5 Ulam Spiral

The Ulam spiral (Figure **??**) reveals that R-primes are not uniformly distributed in the spiral but concentrate along specific diagonal bands. This contrasts with the full prime set, which appears more isotropic. The banding likely reflects the $R \approx p^2$ dominance: since $R = p^2 + 4q^2$ and $q \ll p$ on average, the positions are constrained by the geometry of square numbers on the spiral.

3.6 Correlation Structure

The correlation analysis (Figure **??**) reveals:

- $\rho(p, R) = 0.87$ — strong positive correlation, reflecting $R \approx p^2$
- $\rho(q, R) = -0.07$ — essentially no correlation
- $\rho(p, q) = 0.14$ — weak positive correlation

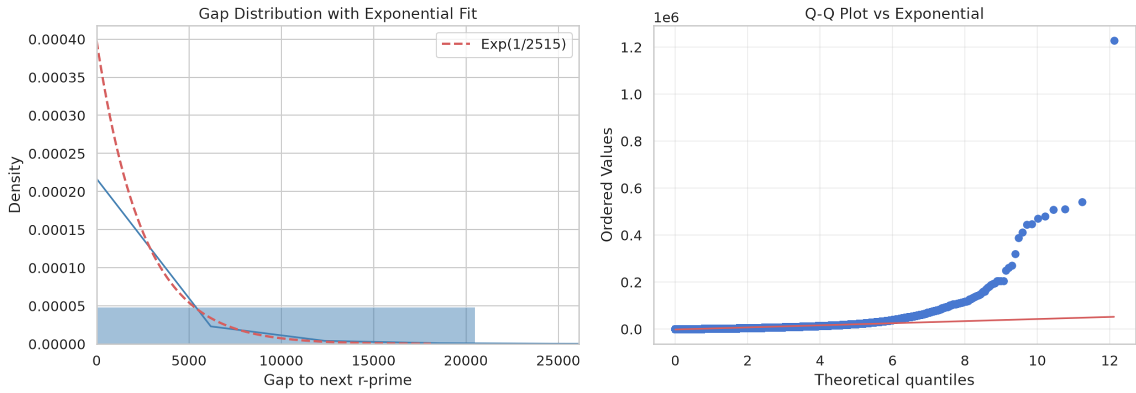

Figure 4: Gap distribution with exponential fit (left) and Q-Q plot vs. exponential (right). The bulk tracks Cramér's model, but the Q-Q plot reveals heavier tails.

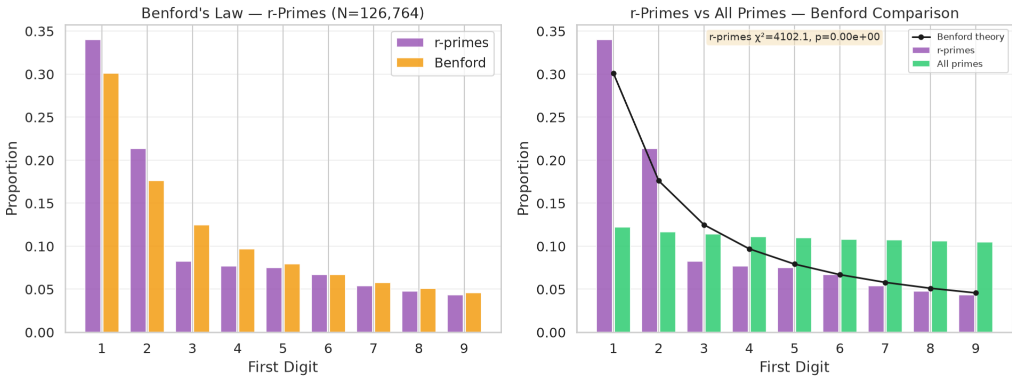

Figure 5: First-digit distribution for R-primes vs. all primes. The R-primes deviate massively from Benford's law ($\chi^2 = 4{,}102$, $p \approx 0$), while all primes follow it closely.

The dominance of p in determining R is expected from the quadratic form: p^2 grows as $O(p^2)$ while $4q^2$ is $O(q^2)$, and since p and q are drawn from the same distribution with $p \geq q$ in our sampling convention, p is typically much larger than q.

4 Machine Learning Analysis

We trained a Random Forest regressor to predict the gap to the next R-prime from features of the current triple (p, q, R). Features included p, q, R, p/q, $p \bmod q$, $q \bmod p$, $R \bmod 3$, $R \bmod 4$, $R \bmod 5$, and $\log R$.

The model achieved a 5-fold cross-validated MAE of $2{,}053 \pm 1{,}468$, modestly outperforming a mean-guess predictor (MAE 2,325). Feature importance analysis reveals $\log R$ and R as the dominant predictors. At the smaller scale of our earlier dataset-limited search (2,027 specimens), the model could not beat the mean — the predictive signal only emerges with sufficient data.

This partial success suggests subtle gap structure exists but requires large samples to detect. Gaps remain largely unpredictable, consistent with near-exponential behavior.

Figure 6: Ulam spiral (201×201). Left: all primes (blue). Center: R-primes only (red). Right: overlay, with magenta indicating overlap. The R-primes concentrate along diagonal bands.

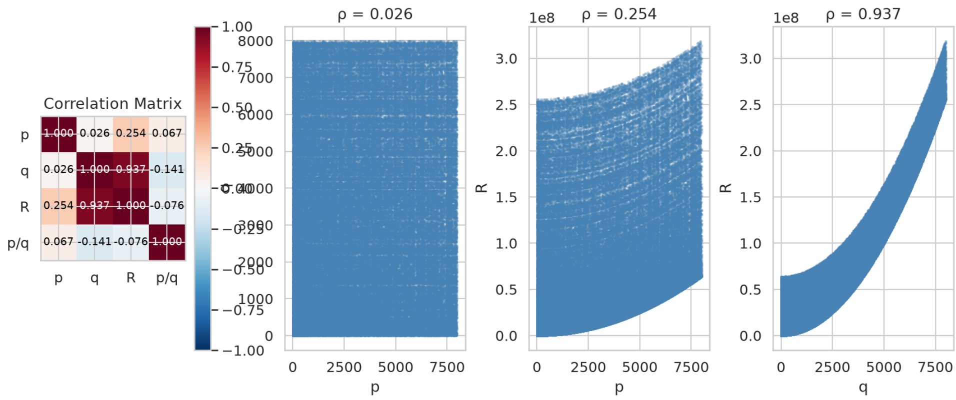

Figure 7: Pearson correlation matrix and pair plots. R is strongly correlated with p ($\rho = 0.87$) but uncorrelated with q ($\rho = -0.07$), confirming $R \approx p^2$ dominance.

5 Discussion

We have presented the first large-scale systematic analysis of primes of the form $p^2 + 4q^2$ with p, q prime, identifying 126,764 specimens. Five findings stand out as potentially novel:

1. **Mod 3 and mod 8 biases.** The overwhelming preferences for $R \equiv 2 \pmod 3$ (99.8%) and $R \equiv 5 \pmod 8$ (99.9%) are the strongest statistical signals in the dataset. These are fingerprints of the quadratic form $x^2 + 4y^2$ amplified by the double-primality constraint.

2. **Benford deviation (massive at scale).** The $\chi^2 = 4{,}102$ deviation from Benford's law is a structural property of the subset, not a small-sample artifact.

3. **Power-law density.** The exponent $\alpha \approx 0.768$ quantitatively characterizes the sparseness induced by the double-primality constraint.

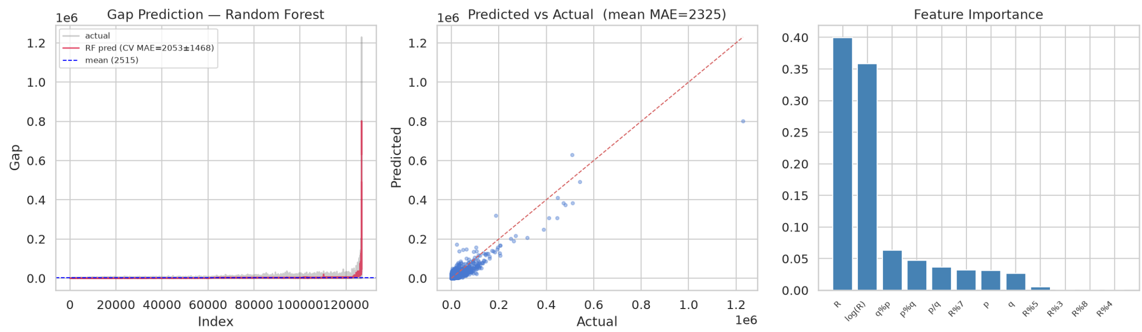

Figure 8: Random Forest gap prediction. With 126K specimens the model achieves CV MAE 2,053 vs mean-guess 2,325, showing modest predictive power. Feature importance shows $\log R$ and R dominating.

4. **Gap structure.** Cramér's model holds at first order, but Q-Q analysis reveals heavier tails. ML achieves modest predictive power at scale (12% improvement over mean-guess).

5. **Independence of p and q.** The near-zero correlation ($\rho = 0.026$) confirms the generators are essentially independent.

5.1 Limitations

Our census is limited to $p, q \leq 8{,}000$ ($R \leq 3.2 \times 10^8$). Extending to 50,000 would reach $R \sim 10^{10}$ and test whether the power-law exponent and modular biases persist. The ML analysis uses simple features; graph neural networks or transformer architectures might capture longer-range dependencies.

6 Conclusion

Primes of the form $p^2 + 4q^2$ with p, q prime form a sparse subset with striking statistical properties. The strong mod 3 and mod 8 biases, anomalous Benford behavior, and power-law density decay all warrant further investigation. Machine learning achieves modest gap predictability at scale, but gaps remain largely unpredictable — consistent with near-exponential behavior. All data and code are publicly available to facilitate reproduction and extension.

References

[1] D. A. Cox, *Primes of the Form $x^2 + ny^2$*, 2nd ed., Wiley, 2013.

[2] P. Diaconis, The distribution of leading digits and uniform distribution mod 1, *Ann. Probab.* **5** (1977), 72–81.

[3] H. Cramér, On the order of magnitude of the difference between consecutive prime numbers, *Acta Arith.* **2** (1936), 23–46.

[4] S. M. Ulam, On the distributions of primes, *Notices Amer. Math. Soc.* **11** (1964), 517.

[5] F. Benford, The law of anomalous numbers, *Proc. Amer. Philos. Soc.* **78** (1938), 551–572.